

# Linear Independence of Variables in INTOXICATE-ER

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## Relationship Between Gender and Xenobiotic

Table 1: **No association between gender and type of xenobiotic.** Cells denote case count (row percentage). F, biological female. M, male. See X for description of exposure categories.

|                        | Biological Sex |          | Total      | p-value | Row vs Rest p | Row vs Rest p (adj) |
|------------------------|----------------|----------|------------|---------|---------------|---------------------|
|                        | F              | M        |            |         |               |                     |
| Exposure Category      |                |          |            |         |               |                     |
| <i>Combination</i>     | 20 (41%)       | 29 (59%) | 49 (100%)  |         | 0.266         | 0.531               |
| <i>Unknown</i>         | 13 (50%)       | 13 (50%) | 26 (100%)  |         | 1.000         | 1.000               |
| <i>Analgesic</i>       | 16 (64%)       | 9 (36%)  | 25 (100%)  |         | 0.153         | 0.407               |
| <i>Antidepressants</i> | 17 (74%)       | 6 (26%)  | 23 (100%)  |         | 0.018         | 0.146               |
| <i>Street Drugs</i>    | 9 (39%)        | 14 (61%) | 23 (100%)  |         | 0.450         | 0.545               |
| <i>CO, As, CN</i>      | 7 (32%)        | 15 (68%) | 22 (100%)  |         | 0.145         | 0.407               |
| <i>Sedatives</i>       | 7 (64%)        | 4 (36%)  | 11 (100%)  |         | 0.476         | 0.545               |
| <i>Alcohol</i>         | 3 (30%)        | 7 (70%)  | 10 (100%)  |         | 0.374         | 0.545               |
| Total                  | 92 (49%)       | 97 (51%) | 189 (100%) |         |               |                     |

<sup>1</sup> NA

```
##
## Call:
## glm(formula = `Dichotomous Outcome` ~ `INTOXICATE SCORE` + Gender,
##      family = binomial, data = train_data)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    -3.10696    0.66253  -4.690 2.74e-06 ***
## `INTOXICATE SCORE`  0.10510    0.04316   2.435  0.0149 *
## GenderM          1.31201    0.61853   2.121  0.0339 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##    Null deviance: 83.702  on 81  degrees of freedom
## Residual deviance: 70.138  on 79  degrees of freedom
## (19 observations deleted due to missingness)
## AIC: 76.138
##
## Number of Fisher Scoring iterations: 5
```

| ##                    |  | 2.5 %       | 97.5 %     |
|-----------------------|--|-------------|------------|
| ## (Intercept)        |  | -4.56486943 | -1.9306520 |
| ## `INTOXICATE SCORE` |  | 0.02486071  | 0.1967328  |
| ## GenderM            |  | 0.13286220  | 2.6000521  |

The sensitivity and specificity of the model on the training data are 0.706 and 0.508, respectively. On the test data, the sensitivity and specificity are 0.267 and 1, respectively.